

TEAM & SELECTED TRACK

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SELECTED TRACK & PROBLEM STATEMENT

Track 2 — Monsoon, Roads & Civic Infrastructure Intelligence. During monsoon events, civic teams lack continuous, prioritized visual intelligence on where road and drainage infrastructure is failing, how severe each issue is, and which intervention should happen first.

CivicPulse converts road and aerial video into evidence-backed civic incidents and connects detection to a verifiable maintenance workflow.

TARGET INCIDENT & CIVIC RELEVANCE

Flagship: Waterlogging. The initial focus is **high-traffic junctions and internal road corridors** in the Electronics City context. The objective is not merely to detect water, but to estimate its spatial extent, persistence and likely operational impact.

Waterlogging is a strong first target because the same visual evidence supports **affected-area estimation, persistence analysis, road-obstruction assessment, severity scoring and maintenance prioritization**. The system complements field inspection rather than replacing it.

SCOPE

PRIMARY	SECONDARY	FUTURE
Waterlogging Affected-area estimation Persistence analysis Road obstruction Severity & priority	Potholes Road-surface maintenance Mask geometry Confidence & context	Drainage Overflow Where sufficient visual evidence exists for reliable detection

INPUT → EVIDENCE → OPERATIONAL OUTPUT

- **Input:** challenge-provided road/aerial video; locally captured or permitted public footage can support development. Precise GPS is not assumed when unavailable.
- **Evidence:** segmentation mask / detection, representative frame or clip, timestamp, source/zone and model confidence.
- **Operational output:** incident class, **severity (0–10), priority (P1/P2/P3), recommended action**, owner and workflow status.

CORE CIVIC WORKFLOW

VIDEO → DETECT / SEGMENT → TEMPORAL ASSOCIATION → INCIDENT → EVIDENCE → SEVERITY → PRIORITY → ACTION → VERIFY → CLOSE

WHY THIS APPROACH

- **Unified segmentation:** one **YOLO26-Seg** pipeline is proposed for both waterlogging and potholes, reducing engineering complexity.
- **Geometry-aware severity:** masks provide the affected shape, enabling coverage-based waterlogging reasoning and geometry-based pothole assessment.
- **Temporal intelligence:** repeated frame detections become one persistent civic incident, reducing duplicate alerts.
- **Actionability:** the output is evidence + severity + priority + recommended action, not just a model prediction.

VALUE PROPOSITION - CivicPulse turns visual monsoon observations into **prioritized, evidence-backed maintenance actions** — helping civic teams detect earlier, decide faster and verify closure.

AI PIPELINE — PROPOSED TECHNICAL DESIGN

1. **Video ingestion:** decode road/aerial video, sample frames and normalize input for inference.
2. **Unified perception:** fine-tune **YOLO26-Seg** with two classes — **waterlogging and pothole**.
3. **Temporal association:** associate masks across frames, suppress duplicates, estimate persistence and select representative evidence.
4. **Incident engine:** convert frame-level predictions into one structured incident with start/end time, evidence, class and source/zone context.
5. **Severity:** water extent/coverage, persistence, obstruction and road criticality; potholes use mask geometry, visibility and context. Scores are normalized to 0–10.
6. **Priority:** combine severity with operational context to **rank P1 / P2 / P3**.
7. **Human verification:** borderline or high-impact incidents require operator confirmation before assignment.

SYSTEM ARCHITECTURE

ROAD / AERIAL VIDEO → FRAME SAMPLING → YOLO26-Seg → TEMPORAL ASSOCIATION → INCIDENT ENGINE

INCIDENT ENGINE → EVIDENCE + CONTEXT → SEVERITY → PRIORITY ENGINE → OPERATOR DASHBOARD

DASHBOARD → VERIFY → ASSIGN → CIVIC ACTION → RE - INSPECTION → CLOSED

SEVERITY & PRIORITY LOGIC

CONFIDENCE	SEVERITY	PRIORITY
How certain is the model that the event exists?	How serious is the physical condition?	How urgently should the team respond?

Initial severity: water extent + persistence + road obstruction + road criticality → severity score. **Priority:** severity + operational context → P1 / P2 / P3. Thresholds and weights will be calibrated against validation data.

ANNOTATION, TRAINING & EDGE STRATEGY

- **Training:** start from COCO-pretrained YOLO26-Seg weights and fine-tune after dataset reconnaissance.
- **Annotation accelerator:** use SAM-assisted mask annotation during dataset preparation; SAM is not part of the runtime pipeline.
- **Primary build:** evaluate YOLO26s-seg for accuracy on the development GPU.
- **Edge path:** benchmark YOLO26n-seg as a smaller deployment candidate and compare accuracy against measured latency and memory use.
- **Deployment principle:** report measured end-to-end performance rather than making an unsupported edge-readiness claim.

TESTING & VALIDATION

- **Visual cases:** heavy rain, wet-road reflections, low light, partial occlusion, vehicle obstruction, low-contrast waterlogging and partially submerged potholes.
- **Model metrics:** precision, recall, F1 and mask mAP / IoU where applicable.
- **Incident metrics:** incident-level recall, false alerts/hour, duplicate-incident rate, evidence completeness and detection-to-alert latency.
- **Failure checks:** reflections mistaken for water; water hidden behind vehicles; fragmented masks; duplicate incidents; unstable adjacent-frame detections.

DASHBOARD, GITHUB & DEMO

- **Dashboard:** incident queue + map/context + evidence frame/clip + confidence + severity + priority + recommended action + owner + status.
- **Workflow:** Detected → Verified → Assigned → In Progress → Re-inspection → Closed.
- **GitHub:** README, reproducible environment, inference entry point, configuration, evaluation scripts, sample outputs and architecture documentation.
- **Demo:** video → segmentation → persistent incident → evidence → severity/priority → assignment → closure.

RISKS & MITIGATION

Wet-surface false positives — temporal persistence + contextual thresholds + human verification. **Occlusion** — multi-frame association + evidence review.

Missing GPS — source/zone identifiers; never fabricate coordinates. **Domain shift** — held-out viewpoints + retraining/fine-tuning workflow.

OBSERVE → UNDERSTAND → PRIORITIZE → ACT → VERIFY